Marvin Durango Cogollo, Ph.D.

Digital Twin and Simulation Engineer

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SUMMARY

Digital twin, simulation and industrial automation engineer (Ph.D., Purdue; R&D at Hitachi America) with 8+ years of experience building digital twins that run next to real machines. My work connects every layer a twin needs: physics-based models of the machine (CFD, FEA, dynamic systems), the hardware and automation that keep it synchronized with the physical asset (sensors, PLCs, robots, OPC UA, MQTT, ROS), and the people who use it (cloud applications and VR/MR interfaces in Unity), now extended with AI through physics-informed neural networks and surrogate models. Delivered a hydraulic-machine digital twin validated within 7% tracking error and collaborative-robot workflows that improved capacity predictability by 89%.

TECHNICAL SKILLS

Modeling and Simulation: Finite element analysis (FEA, Ansys Mechanical), computational fluid dynamics (CFD) with mesh-based (Ansys Fluent, Ansys CFX) and particle-based (SPH, CFD-DEM) solvers, multiphysics simulation, discrete-event simulation (Visual Components)

Controls and System Modeling: MATLAB/Simulink, physics-based dynamic modeling, lumped-parameter models, control system design, system identification

Digital Twins: Physical-digital synchronization, virtual commissioning, sensor fusion, prediction and system awareness using physics-based, data-driven (AI, LLMs) and hybrid approaches (physics-informed neural networks, surrogate models), verification and validation (V&V), design of experiments (DOE)

Industrial Automation and Robotics: Industrial automation, process and test automation, controls integration, PLC programming, equipment commissioning, robot programming (UR5, UR10e collaborative robots), ROS, inverse kinematics for serial and hydraulic manipulators, motion planning, spatial awareness and computer vision with LiDAR and SLAM

Industrial IoT and Data: OPC UA, MQTT and industrial communication protocols, cloud processing (AWS), end-to-end application development (REST APIs, web dashboards), data acquisition and signal conditioning, edge computing

CAD and Design: 3D CAD modeling of mechanical components and assemblies (Autodesk Inventor, Siemens NX), test-bench and fixture design

Software and Programming: Python (NumPy, SciPy, Flask, asyncio), C++, C#, MATLAB, Docker, Git, Linux

3D, XR and GPU: User interface (UI) design for 3D and immersive applications, Unity, virtual reality (Meta Quest, Oculus Rift), mixed reality (HoloLens 2), NVIDIA Isaac Sim, NVIDIA Omniverse/PhysX (familiar), OpenGL, GLSL compute shaders, particle animation

AI and Machine Learning: Physics-informed neural networks (PINNs; inverse PINN for hydraulic state estimation, in progress), Bayesian regression, computer vision (YOLO, OpenCV, CNNs)

PROFESSIONAL EXPERIENCE

Graduate Researcher, Digital Twins and Simulation — Aug 2021 – Present
Purdue University, School of Engineering Technology — West Lafayette, IN

- Designed and built an instrumented two-joint hydraulic crane and its real-time digital twin end to end: hydraulic power unit, proportional valves, cylinder, hydraulic motor, electro-hydrostatic actuator, pressure, position, torque and encoder sensors, data acquisition and edge computing; components modeled in CAD (Inventor, NX); the twin tracks the physical machine within **7% error**.
- Developed physics-based models of hydraulic cylinders, motors, electro-hydrostatic actuators, hydrostatic transmissions and proportional valves; ported them from MATLAB to open-source Python with $<$**1.3% NRMSE**, enabling license-free deployment on low-cost edge devices.
- Built a Unity/HoloLens 2 mixed-reality operator interface in which hand-defined targets drive inverse kinematics (FABRIK) and collision-free path planning (RRT*), streamed over TCP/IP to the physical boom and swing.
- Automated the test system end to end: remote start-up and shutdown of the hydraulic power unit, software speed control, automated operating sequences with fault handling, and unattended test campaigns, reducing manual operation and making testing repeatable.
- Building a six-station production-line digital twin for Purdue's Smart Learning Factory in Visual Components, synchronized with the physical cell over OPC UA (**2 of 6 stations** recreated for virtual commissioning) and extending it to NVIDIA Isaac Sim; validated UR10e kinematics against the physical robot.
- Connected shop-floor equipment to cloud processing on AWS and integrated a process line over MQTT in Purdue's IoT laboratories, streaming real-time machine data into simulation and monitoring.

R&D Intern, Smart and Digital Manufacturing Jun 2023 – Jan 2024
Hitachi America, Industrial Innovation Laboratory Farmington Hills, MI

- Programmed UR5 collaborative-robot tasks for human-robot workflows, improving capacity predictability by **89%** and reducing operator fatigue by **more than 50%**.
- Built a sensor-integrated digital twin of a manufacturing cell and streamed live robot data through ROS, so robot programs and layout changes were validated before physical commissioning.
- Developed discrete-event simulations of packaging processes and demand scenarios to evaluate throughput and capacity before investment.
- Performed market and patent analysis of robotics and digital-twin technologies to identify customer-driven opportunities.

SELECTED PROJECTS

- **GPU physics engine (C++, OpenGL, GLSL):** real-time solver in compute shaders, including SPH fluids with **144,000+ particles**, XPBD constraints and continuum-mechanics beams (Green strain tensor), coupled to interactive 3D rendering.
- **Hydraulic component CFD and visualization (Ansys CFX, Unity):** modeled gear-pump and pressure-relief-valve flow and rendered internal velocity and pressure fields in virtual and mixed reality.
- **Multiphase process modeling (Ansys Fluent, CFD-DEM):** optimized hydrocyclone separators using coupled fluid-particle and Eulerian multiphase models; benchmarked RNG k-epsilon, RSM and LES turbulence models; validated inputs with laser-diffraction particle sizing (*Fluids*, 2020).
- **Meshless CFD for moving parts (SPH):** simulated a hydraulic spool valve without remeshing (ASME/BATH FPMC 2023) and authored a review of SPH for internal flow (*Journal of Hydrodynamics*, accepted).
- **Pneumatic digital twin:** hybrid physics/data-driven estimator (EKF, LuGre friction, Bayesian ridge regression) with mixed-reality control and an open dataset (journal manuscript in revision).

LEADERSHIP AND TEACHING

- Designed and taught a project-based smart-manufacturing course (Fall 2025) in which students build digital twins of a production line with robots, conveyors and AGVs; instructor or TA for fluid-power and pneumatic-control courses (2021 – present).
- Mentored two undergraduate interns (industrial connectivity; LiDAR object and distance detection) and two undergraduate researchers who co-authored a journal manuscript; peer reviewer for the Maha Fluid Power Conference.

EDUCATION

Ph.D., Engineering Technology Dec 2026
Purdue University — dissertation defended Sept 2026: physics-based digital twins for hydraulic machinery West Lafayette, IN
M.S., Mechanical Engineering 2019
Universidad de los Andes — thesis: CFD-DEM optimization of hydrocyclones Bogota, Colombia
B.S., Mechanical Engineering 2015
Universidad Tecnologica de Bolivar Cartagena, Colombia

SELECTED PUBLICATIONS

- Durango, M., & Garcia-Bravo, J. M. (2026). Review of SPH for internal flow in hydraulic components. *Journal of Hydrodynamics*, accepted.
- Durango, M., & Garcia-Bravo, J. M. (2024). Physics-based digital twin for hydraulic equipment. *DigiTwin Conference*.
- Durango, M., Garcia-Bravo, J., Newell, B., & Gonzalez-Mancera, A. (2020). CFD modeling of hydrocyclones. *Fluids*, 5(3), 118.

Languages: English and Spanish (bilingual).